

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – X

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Class: TYIT	Batch: 1
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Data Preparation and Transformation using Tableau Prep

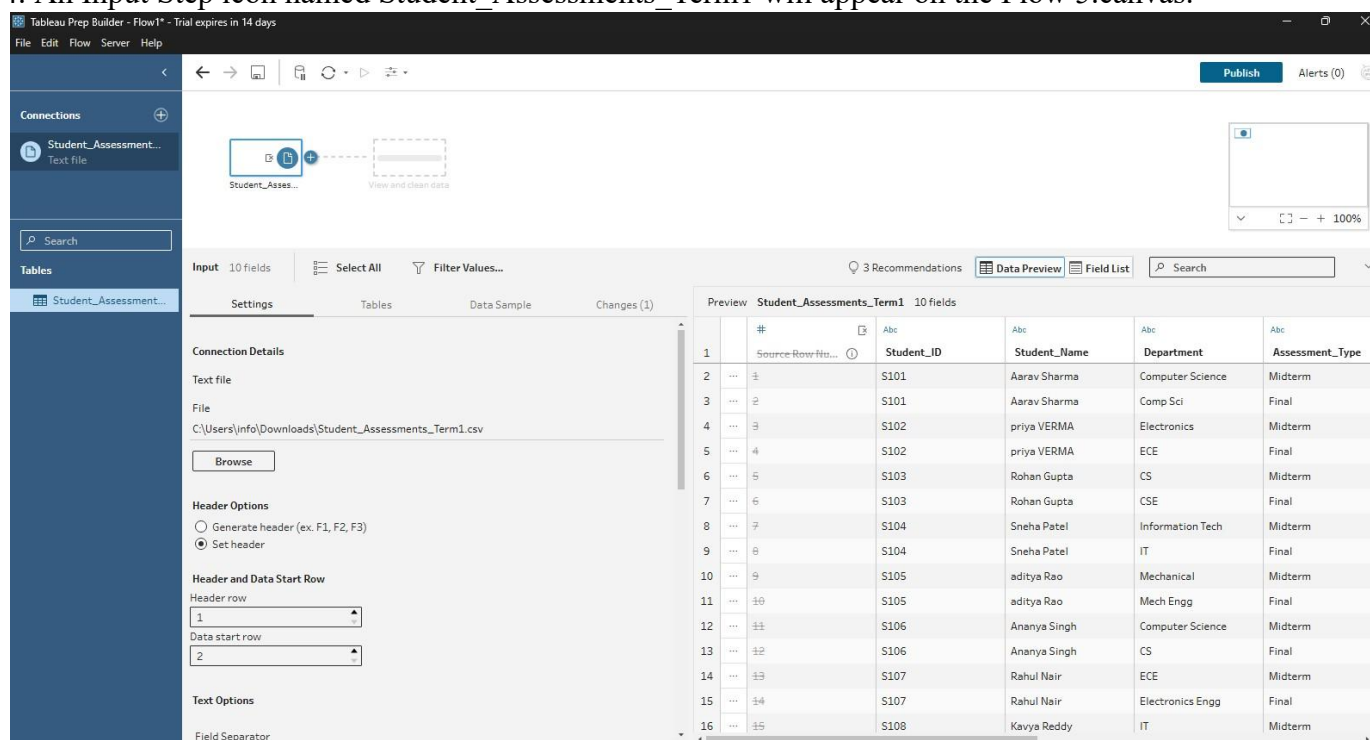
A) Connect and load data into Tableau Prep Step

1: Open Tableau Prep Builder

1. Launch Tableau Prep Builder on your machine.
2. On the start screen, locate the Connections pane on the left-hand side.

Step 2: Load the First Dataset (Student_Assessments_Term1.csv)

1. In the Connections pane, click the + (Add connection) button.
2. In the list of file formats, select Text file.
3. Browse to the folder where you saved the CSVs, select Student_Assessments_Term1.csv, and click Open.
4. An Input Step icon named Student_Assessments_Term1 will appear on the Flow canvas.



5. In the bottom Input configuration pane, verify:
 - o Field Separator: Comma (,)
 - o First row contains header: Checked

Step 3: Load the Second and Third Datasets

1. In the top-left Connections pane, click the + (Add connection) button again.
2. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open.

3. Click + (Add connection) once more -> select Text file - >choose Course_Credit_Master.csv -> click Open.
4. You will now see three separate Input icons on your Flow canvas.

Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon.
2. Click the small + icon that appears to its right.
3. Select Clean Step (the step with the cleaning tool icon).

The screenshot shows the Alteryx interface with the 'Add' menu open over the 'Course_Credit_Master' input icon. The menu includes options like 'Clean Step', 'New Rows', 'Aggregate', 'Pivot', 'Join', 'Union', 'Script', 'Prediction', 'Output', and 'Insert Flow'. The 'Clean Step' option is highlighted. The flow canvas shows three input icons: Student_Assessments_Term1, Student_Assessments_Term2, and Course_Credit_Master. The 'Course_Credit_Master' input is highlighted, and the 'Add' menu is open, showing options like 'Clean Step', 'New Rows', 'Aggregate', 'Pivot', 'Join', 'Union', 'Script', 'Prediction', 'Output', and 'Insert Flow'.

4. The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.

The screenshot shows the Alteryx interface with the 'Clean 1' step connected to the 'Student_Assessments_Term1' input. The 'Profile Pane' is open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1. The 'Clean 1' step is highlighted, and the 'Profile Pane' shows data for columns like Student_ID, Student_Name, Department, Assessment..., Exam_Date, Subject_Code, and Marks_Obtain... The 'Profile Pane' also shows a 'Changes (0)' section with a table of data.

Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject_Code	Marks_Obtain...
S101	Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35
S102	aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55
S103	Ananya Singh	CS		2025 09 15	IT101	75
S104	Kavya Reddy	CSE		2025 11 20	ME101	95
S105	priya VERMA	ECE		2025/09/15		
S106	Rahul Nair	Electronics		2025/09/15		
S107	Rohan Gupta	Electronics Engg		2025/11/20		
S108	Sneha Patel	Info Tech				

B| Clean and profile data Step

1: Inspect the Profile Pane

1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Look at the cards displayed in the Profile Pane:

- o Each card represents one column.
- o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao ').
- o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/, -, .).

Step 2: Clean Whitespace and Standardize Letter Casing

1. On the Student_Name profile card:
 - o Click the More options (...) icon at the top right of the card.
 - o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).
 - o Click More options (...) again --> Clean --> select Make Uppercase (or Make Lowercase / standard proper formatting).

Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks
Aarav Sharma	Comp Sci	Final	15-09-2025	CS101	35	100
Aditya Rao	Computer Science	Midterm	20-11-2025	EC101	55	
Ananya Singh	CS		2025.09.15	IT101	75	
Kavya Reddy	CSE		2025.11.20	ME101	95	
Priya Verma	ECE		2025/09/15			
Rahul Nair	Electronics		2025/11/20			
Rohan Gupta	Electronics Engg					
Rohan Gupta	Info Tech					
Sneha Patel	Information Tech					
	IT					
	Mech Engg					
	Mechanical					

2. The card updates immediately, consolidating duplicate whitespace variations.

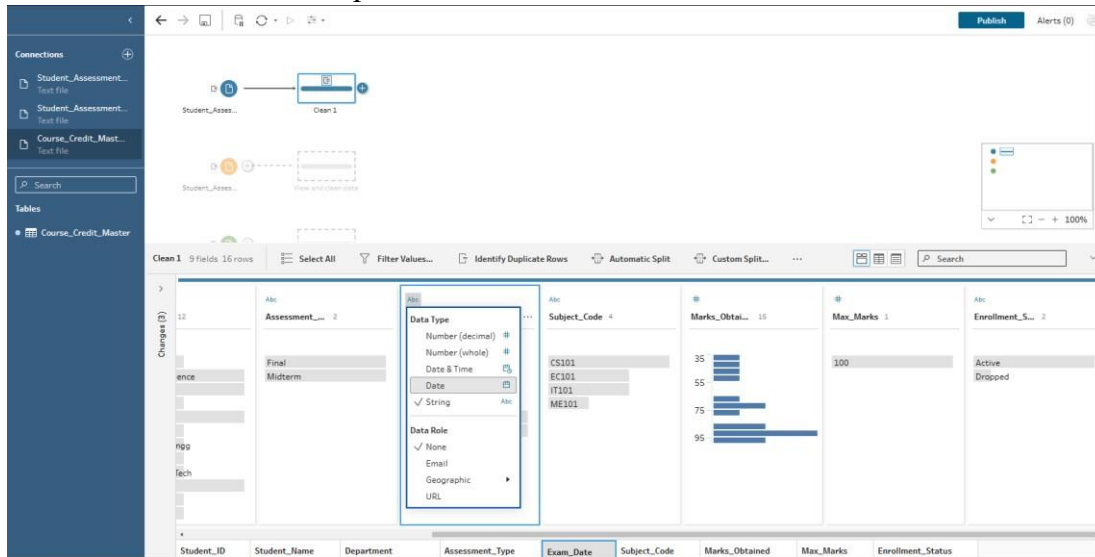
Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks
AARAV SHARMA	Comp Sci	Final	15-09-2025	CS101	35	100
ADITYA RAO	Computer Science	Midterm	20-11-2025	EC101	55	
ANANYA SINGH	CS		2025.09.15	IT101	75	
KAVYA REDDY	CSE		2025.11.20	ME101	95	
PRIYA VERMA	ECE		2025/09/15			
RAHUL NAIR	Electronics		2025/11/20			
ROHAN GUPTA	Electronics Engg					
ROHAN GUPTA	Info Tech					
SNEHA PATEL	Information Tech					
	IT					
	Mech Engg					
	Mechanical					

Step 3: Convert Exam_Date to Date Data Type 1.

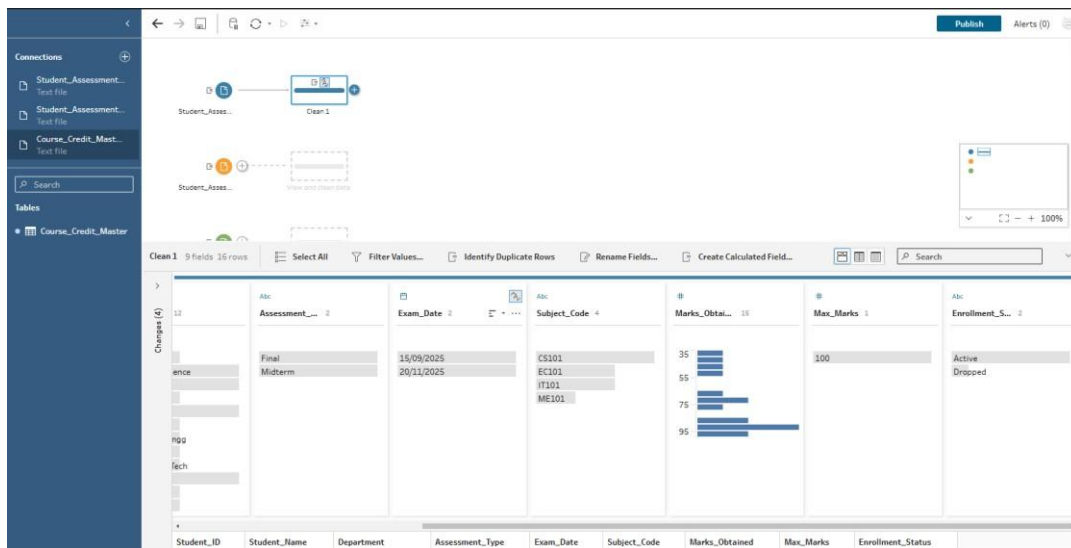
Locate the Exam_Date profile card.

2. Click the data type icon (currently showing Abc) directly above the column name.

3. Select Date from the dropdown menu.



4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.



Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv) 1. On

the Flow canvas, hover over the Student_Assessments_Term2 input icon.

2. Click + --> select Clean Step (creates Clean 2).

3. On the Full_Name card in Clean 2: o Click More options (...) --> Clean --> Trim Spaces.

The screenshot shows the Power BI Desktop interface. On the left, the 'Connections' pane lists 'Student_Assessment...' and 'Course_Credit_Mast...'. The 'Tables' pane shows 'Course_Credit_Master'. The main view displays a data table with columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, and Course_Code. A context menu is open over the 'Full_Name' column, showing options like 'Filter', 'Clean', 'Group Values', 'Split Values', 'Identify Duplicate Rows', 'View State', 'Rename Field', 'Duplicate Field', 'Keep Only Field', 'Create Calculated Field', 'Hide Field', and 'Remove'. A sub-menu is open for 'Clean', showing options like 'Make Uppercase', 'Make Lowercase', 'Make Titlecase', 'Remove Letters', 'Remove Numbers', 'Remove Punctuation', 'Trim Spaces', 'Remove Extra Spaces', and 'Remove All Spaces'.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201
S103	Kavya Reddy	CSE			IT201
S104	Priya Verma	ECE			
S106	Rahul Nair	Electronics			
S107	Rohan Gupta	Information Tech			
S108	Sneha Patel	IT			

4. On the Assessment Date card: Click the Abc icon --> change the data type to **Date**.

The screenshot shows the Power BI Desktop interface. The 'Assessment_Date' column is selected, and a 'Data Type' dropdown menu is open. The menu shows options for 'Data Type' (Number (decimal), Number (whole), Date & Time, Date, String) and 'Data Role' (None, Email, Geographic, URL). The 'Date' option is selected under 'Data Type'. The table below shows the data for the 'Assessment_Date' column.

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201	50		
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201	60		
S103	Kavya Reddy	CSE			IT201	70		
S104	Priya Verma	ECE				75		
S106	Rahul Nair	Electronics				82		
S107	Rohan Gupta	Information Tech				88		
S108	Sneha Patel	IT				90		

Std_ID	Full_Name	Dept	Exam_Type	Assessment...	Course_Code	Score
S101	Aarav Sharma	Computer Science	Final	10/02/2026	CS201	50
S102	Ananya Singh	CS	Midterm	25/04/2026	EC201	60
S103	Kavya Reddy	CSE			IT201	70
S104	Priya Verma	ECE				75
S106	Rahul Nair	Electronics				82
S107	Rohan Gupta	Information Tech				88
S108	Sneha Patel	IT				90
						94
						95
						96
						98
						100

CI Group and standardize values

Step 1: Inspect the Department Field in Term 1 (Clean 1)

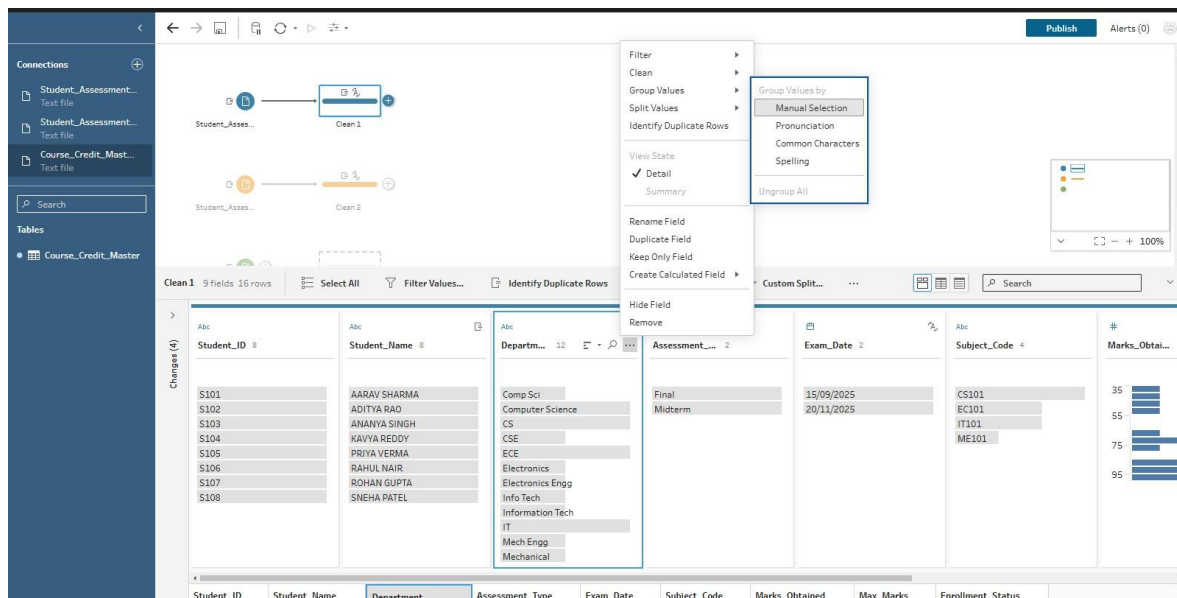
1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Locate the Department card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
 - o Computer Science (Computer Science, Comp Sci, CS, CSE)
 - o Electronics (Electronics, ECE, Electronics Engg)
 - o Information Technology (Information Tech, IT, Info Tech)
 - o Mechanical Engineering (Mechanical, Mech Engg)

Student_ID	Student_Name	Department	Assessment...	Exam_Date	Subject_Code	Marks_Obtain...
S101	AARAV SHARMA	Comp Sci	Final	15/09/2025	CS101	35
S102	ADITYA RAO	Computer Science	Midterm	20/11/2025	EC101	55
S103	ANANYA SINGH	CS			IT101	75
S104	KAVYA REDDY	CSE			ME101	95
S105	PRIYA VERMA	ECE				
S106	RAHUL NAIR	Electronics				
S107	ROHAN GUPTA	Electronics Engg				
S108	ROHAN GUPTA	Info Tech				
	SNEHA PATEL	Information Tech				
		IT				
		Mech Engg				
		Mechanical				

Step 2: Group Values Using Pronunciation / Spelling or Manual Selection 1.

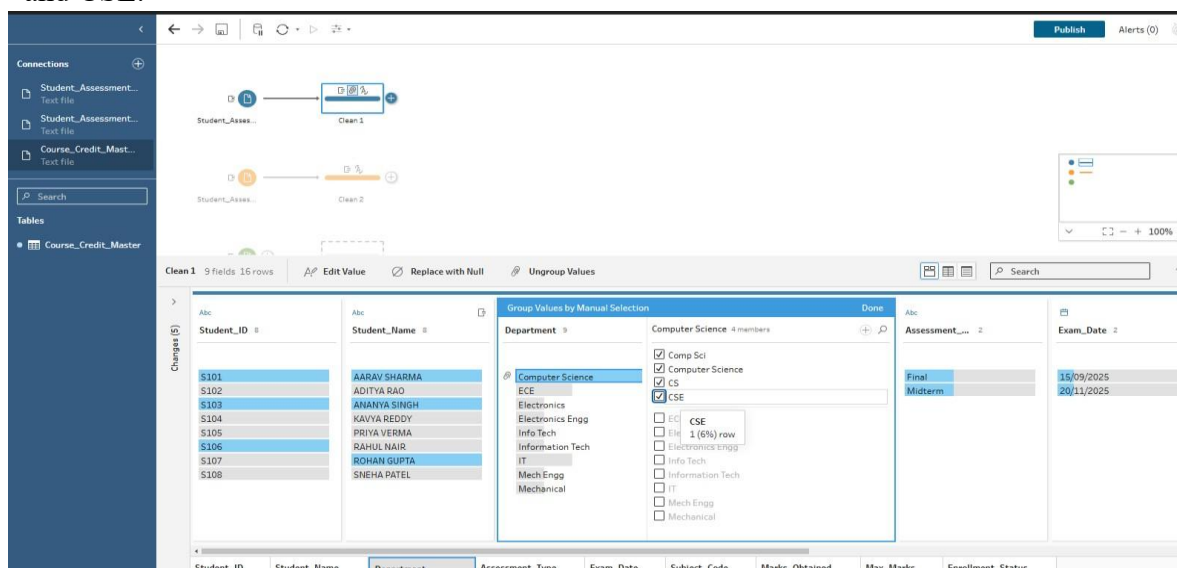
On the Department card, click the More options (...) menu at the top right.

2. Go to Group Values -->select Manual Selection (or Pronunciation / Common Characters).



3. In the Grouping editor:

- o Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE.



- o Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables

- Course_Credit_Master

Clean 1 9 fields 16 rows

Abc Student_ID 8

Abc Student_Name 8

Group Values by Manual Selection

Department 8

Electronics & Communication 2 members

- ☒ Electronics & Communication
- ☒ Electronics Engg
- ☐ Computer Science
- ☐ ECE
- ☐ Electronics
- ☐ Info Tech
- ☐ Information Tech
- ☐ IT
- ☐ Mech Engg
- ☐ Mechanical

Assessment_Type 2

Final

Midterm

Exam_Date 2

15/09/2025

20/11/2025

Student_ID Student_Name Department Assessment_Type Exam_Date Subject_Code Marks_Obtained Max_Marks Enrollment_Status

o Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables

- Course_Credit_Master

Clean 1 9 fields 16 rows

Abc Student_ID 8

Abc Student_Name 8

Group Values by Manual Selection

Department 5

Information Tech 3 members

- ☒ Info Tech
- ☒ Information Tech
- ☒ IT
- ☐ Computer Science
- ☐ ECE
- ☐ Electronics
- ☐ Electronics & Communication
- ☐ Electronics Engg
- ☐ Mech Engg
- ☐ Mechanical

Assessment_Type 2

Final

Midterm

o Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.

Null Ungroup Values

Group Values by Manual Selection

Depa... 4

Mechanical 2 members

- ☒ Mech Engg
- ☒ Mechanical
- ☐ Computer Science
- ☐ ECE
- ☐ Electronics
- ☐ Electronics & Communication
- ☐ Electronics Engg
- ☐ Info Tech
- ☐ Information Tech
- ☐ Information Technology
- ☐ IT

Computer Science

Electronics & Communication

Information Technology

Mechanical Engineering

4. Click Done at the top right of the editor.

5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.

The screenshot displays the Power BI interface. On the left, the 'Connections' pane shows three text files: 'Student_Assessment...', 'Student_Assessment...', and 'Course_Credit_Mast...'. Below this, the 'Tables' pane shows a table named 'Student_Assessment...'. The main area shows a visual of the 'Clean 1' table, which has 9 fields and 16 rows. The table is displayed in a grid view with the following columns: 'Student_ID', 'Student_Name', 'Department...', 'Assessment...', 'Exam_Date', and 'Subject_Co'. The 'Department...' column shows 12 entries, which are grouped into 4 categories: 'Comp Sci', 'Computer science', 'CS', 'CSE', 'ECE', 'Electronics and Communicat', 'Electronics Engg', 'Info Tech', 'Information Technology', 'IT', 'Mech Engg', and 'Mechanical Engineering'. A tooltip for 'Mechanical Engineering' indicates '1 (6%) row'. The 'Assessment...' column shows 'Final' and 'Midterm'. The 'Exam_Date' column shows dates from 15-09-2025 to 2025/11/20. The 'Subject_Co' column shows codes like CS101, EC101, IT101, and ME101. The interface includes a 'Publish' button and 'Alerts (0)' in the top right corner.

Step 3: Repeat Grouping on Term 2 (Clean 2)

1. Click on the Clean 2 step connected to Student_Assessments_Term2.

2. Locate the Dept card.

3. Click More options (...) -->Group Values -->Manual Selection: o Group CS, CSE, and Computer Science into Computer Science.

o Group ECE and Electronics into Electronics. o Group IT and Information Tech into Information Tech.

4. Click Done.

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Search

Tables

- Student_Assessment...

Clean 2 9 fields 14 rows

Keep Only Exclude Edit Value Replace with Null

Std_ID	Full_Name	Dept	Exam_Type
S101	Aarav Sharma	Computer Science	Final
S102	Ananya Singh	Electronics	Midterm
S103	Kavya Reddy	Information Tech	
S104	Priya Verma		
S106	Rahul Nair		
S107	Rohan Gupta		
S108	Sneha Patel		

D| Merge mismatched fields

Step 1: Create the Union Step on the Canvas

1. On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step icon.

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Search

Tables

- Student_Assessment...

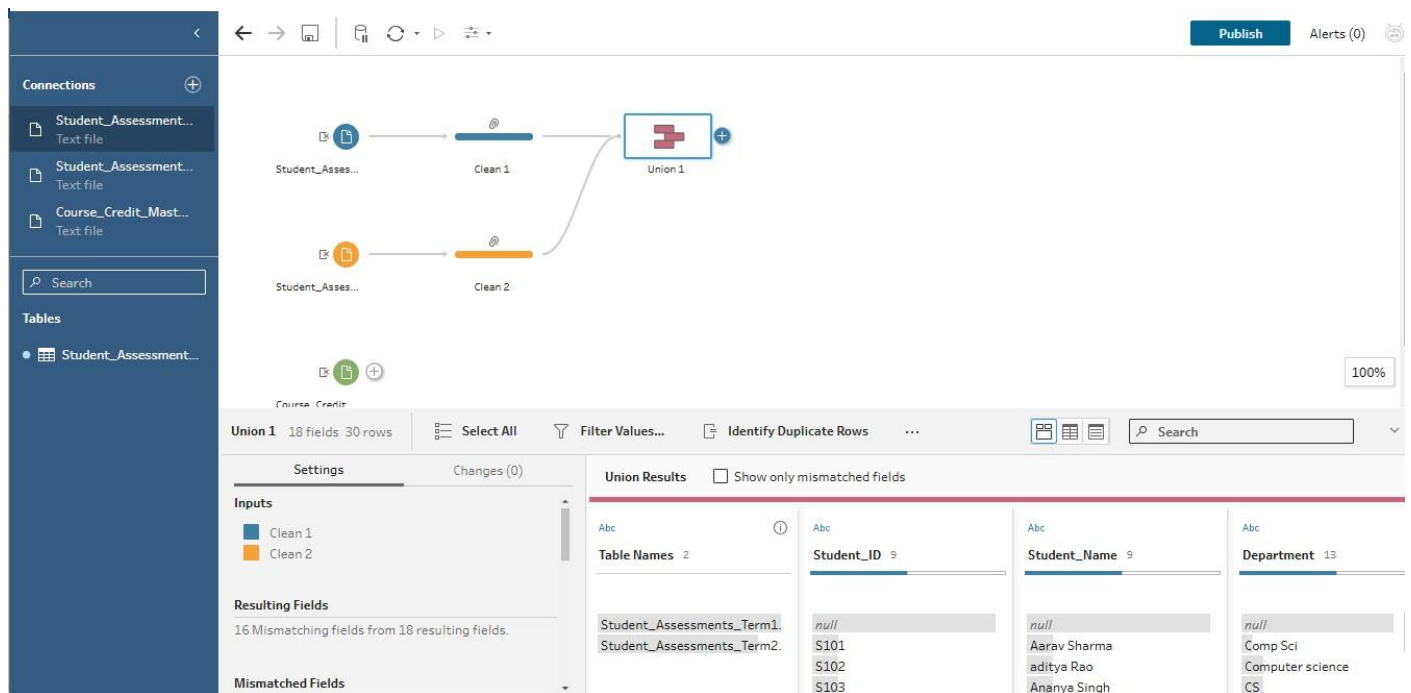
Student_Asses... Clean 2

Student_Asses... Clean 2

Union

Join

2. When the drop zone appears, hover over Union and release the mouse.
3. A new Union 1 step is created, connecting both transformation branches.



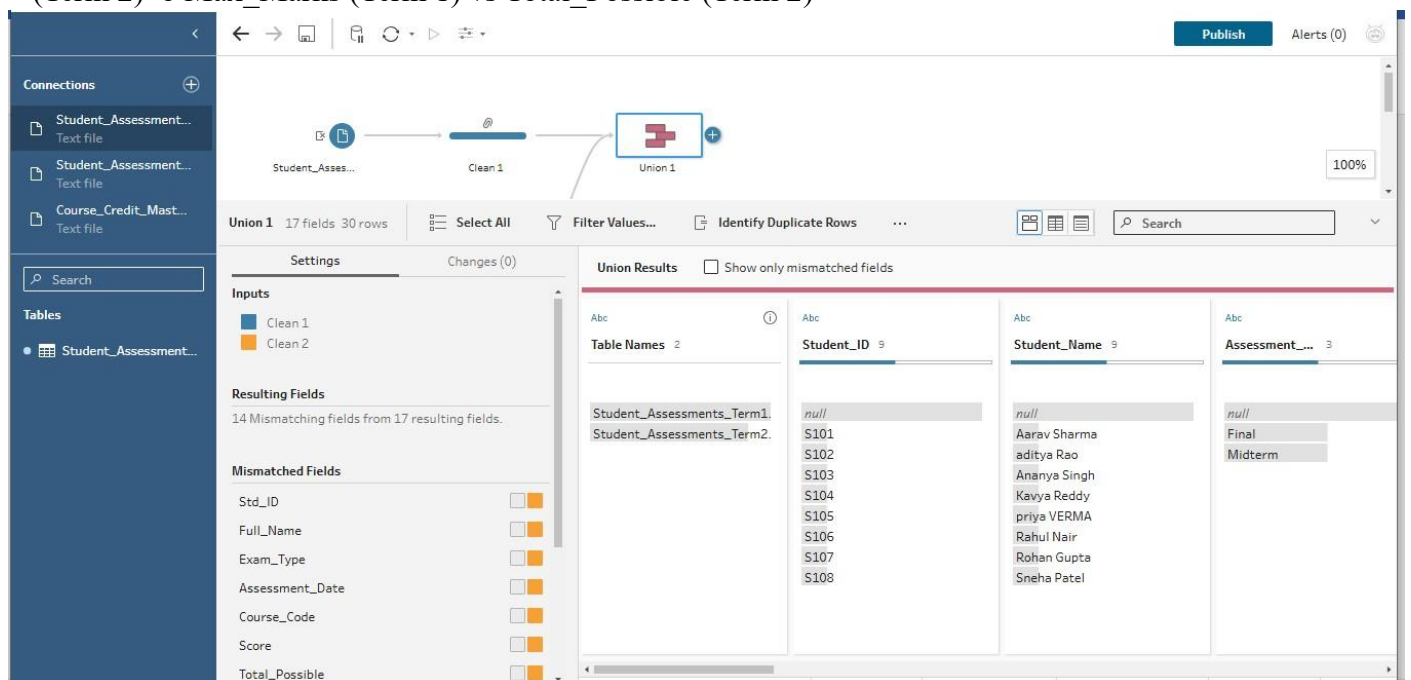
Step 2: Inspect Mismatched Fields in the Union Pane 1.

Click on the Union 1 step.

2. In the bottom Union summary pane, look at the Mismatched Fields section.

3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:

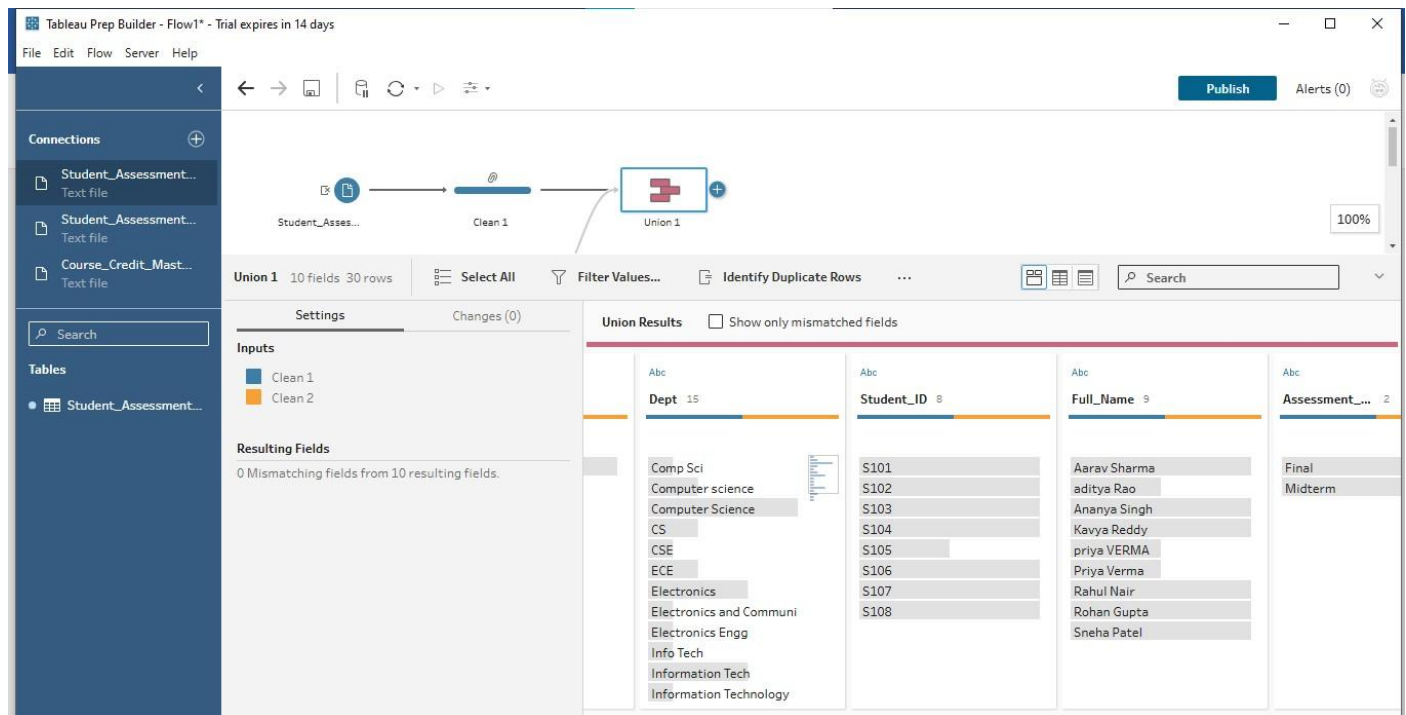
- o Student_ID (Term 1) vs Std_ID (Term 2)
- o Student_Name (Term 1) vs Full_Name (Term 2)
- o Department (Term 1) vs Dept (Term 2)
- o Assessment_Type (Term 1) vs Exam_Type (Term 2)
- o Exam_Date (Term 1) vs Assessment_Date (Term 2)
- o Subject_Code (Term 1) vs Course_Code (Term 2)
- o Marks_Obtained (Term 1) vs Score (Term 2)
- o Max_Marks (Term 1) vs Total_Possible (Term 2)



Step 3: Merge the Mismatched Fields

1. In the Mismatched Fields list (or the union profile pane):

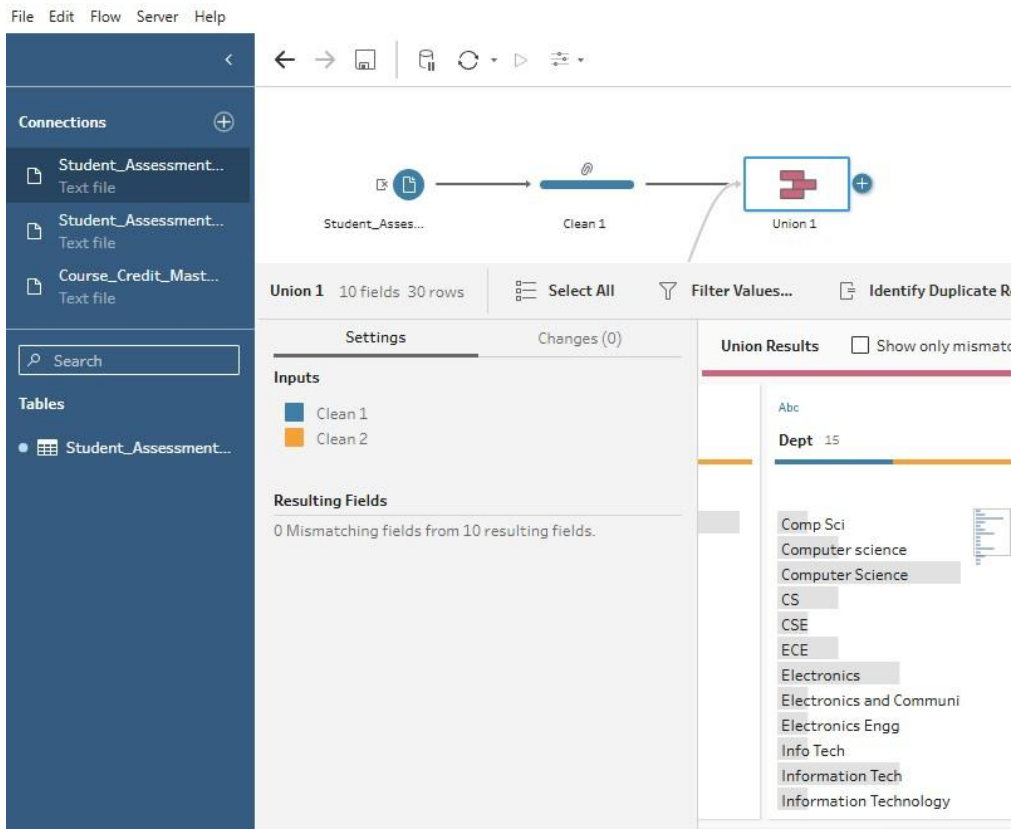
- o Click and drag Std_ID directly on top of Student_ID to merge them into one column.
- o Drag Full_Name onto Student_Name.
- o Drag Dept onto Department.



Note: If You cant drag it Observe in the left pane click the dept column the hover near department column's plus icon and add the dept column into it

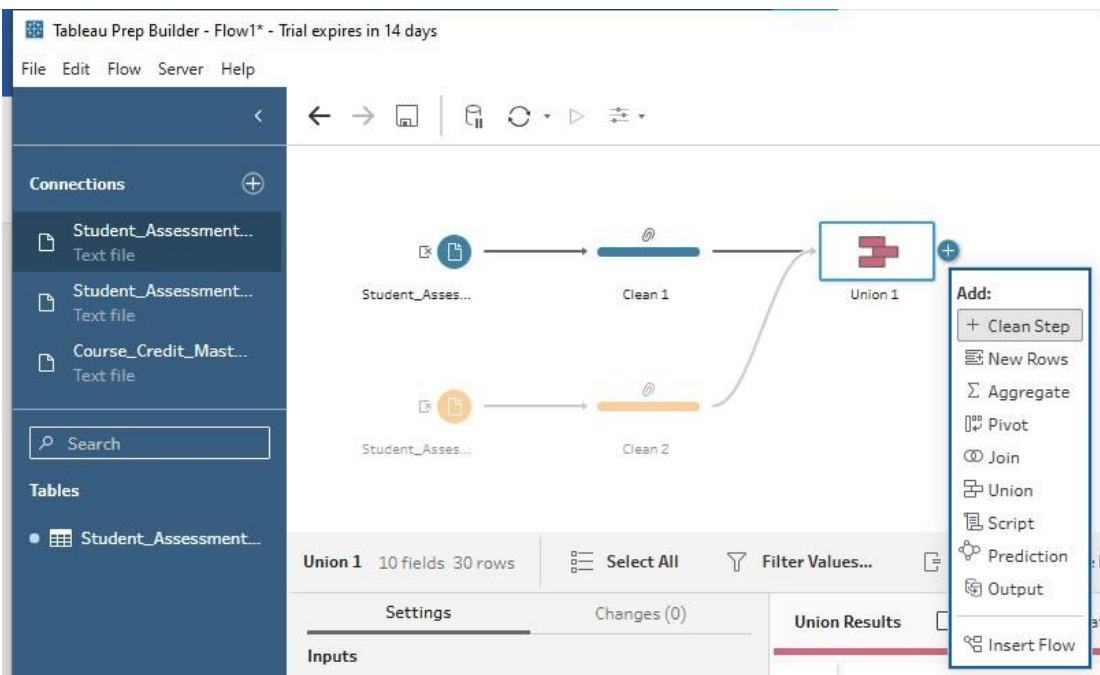
- o Drag Exam_Type onto Assessment_Type.
- o Drag Assessment_Date onto Exam_Date.
- o Drag Course_Code onto Subject_Code.
- o Drag Score onto Marks_Obtained.
- o Drag Total_Possible onto Max_Marks.

2. Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.



Step 4: Add a Clean Step Post-Union

1. Click the + icon next to the Union 1 node.



2. Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.
3. Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

The screenshot shows the Alteryx interface with a data flow diagram and a data preview table.

Data Flow Diagram:

- Two input sources (Student_Asses...) feed into Clean 1 and Clean 2.
- Clean 1 and Clean 2 feed into Union 1.
- Union 1 feeds into Clean 4.

Data Preview Table:

Table Names	Enrollment_S...	Dept	Student_ID	Full_Name	Assessment
Student_Assessments_Term1	Active	Comp Sci	S101	Aarav Sharma	Final
Student_Assessments_Term2	Dropped	Computer science	S102	aditya Rao	Midterm
		Computer Science	S103	Ananya Singh	
		CS	S104	Kavya Reddy	
		CSE	S105	priya VERMA	
		ECE	S106	Priya Verma	
		Electronics	S107	Rahul Nair	
		Electronics and Communi	S108	Rohan Gupta	
		Electronics Engg		Sneha Patel	

E| Perform aggregations in Prep

Step 1: Add a Term Identifier Field in Clean 3

1. Click on the Clean 3 step (the clean step created after Union 1).
2. In the toolbar above the Profile cards, click Create Calculated Field You have to click three dots on the toolbar.
3. Set the Field Name to: Term
4. Enter the formula:

Code snippet

```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```

4.Click Save.

The screenshot shows the 'Add Field' dialog box in Alteryx.

Field Name: Term

Formula:

```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```

Functions List:

- ABS
- ACOS
- AND
- AREA
- ASC
- ASCII
- ASIN
- ATAN
- ATAN2
- AVG
- BUFFER
- CASE
- CEILING
- CHAR
- CONTAINS
- COS

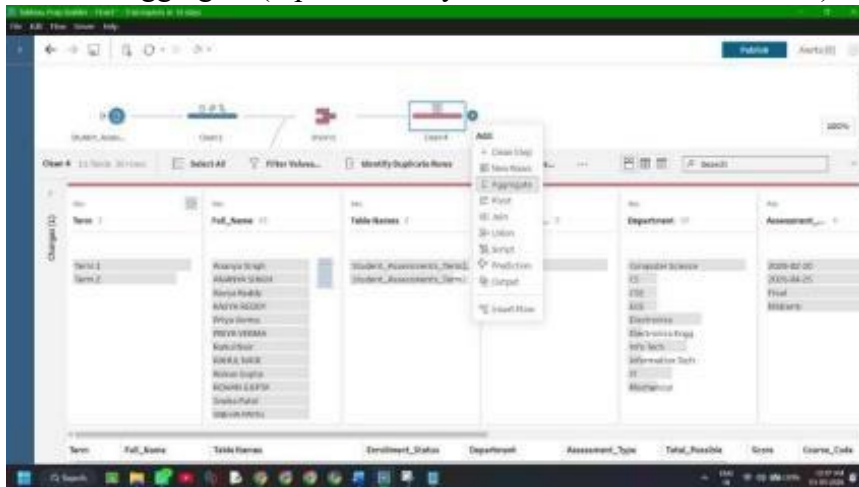
ABS(number)
Returns the absolute value of the given number.
Example: ABS(-7)

Step 2: Add an Aggregate Step to the Flow Canvas 1.

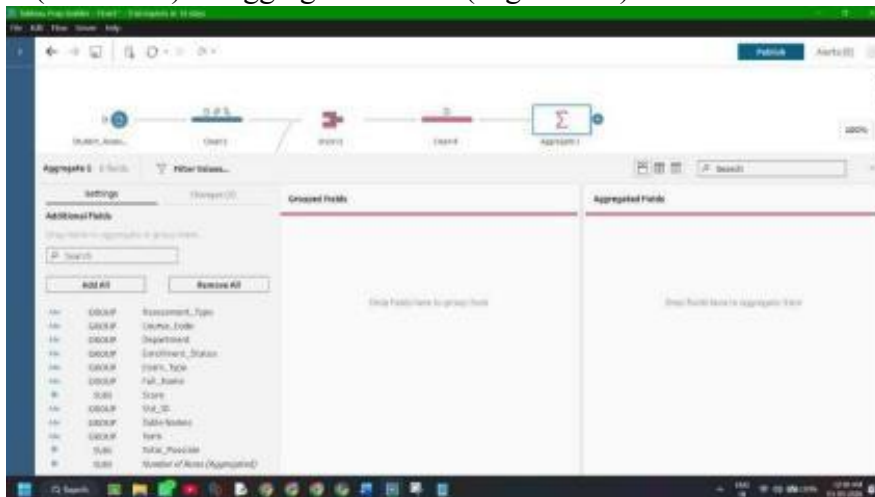
Hover over the Clean 3 node on the Flow canvas.

2. Click the + icon next to it.

3. Select Aggregate (represented by the summation icon Σ).



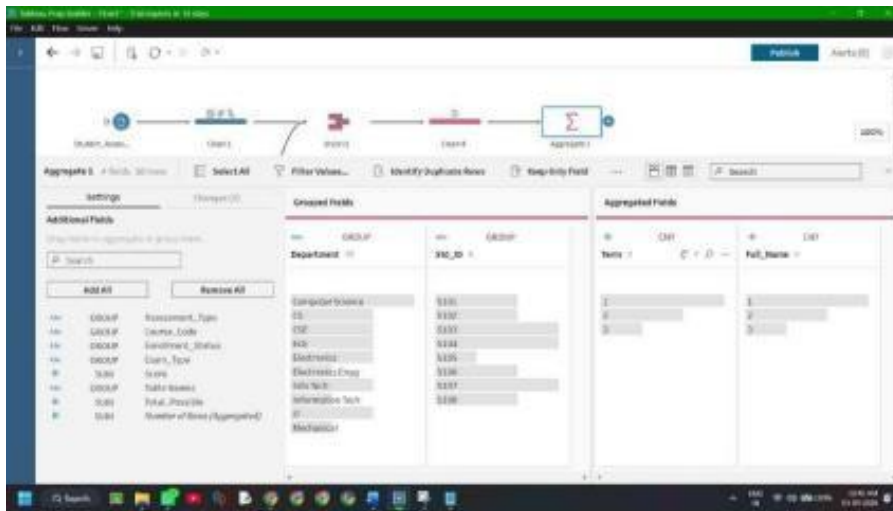
4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane: o Grouped Fields (Left Zone) o Aggregated Fields (Right Zone)



Step 3: Configure Grouped Fields (Dimensions)

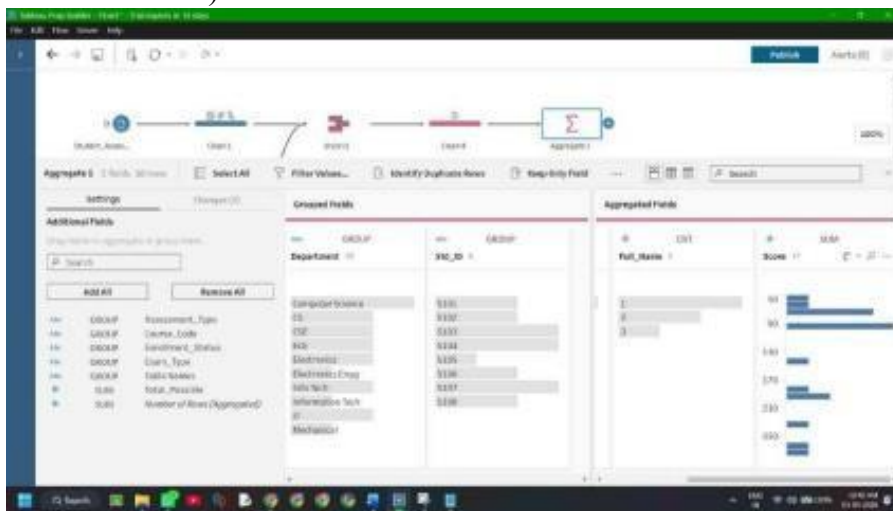
From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.

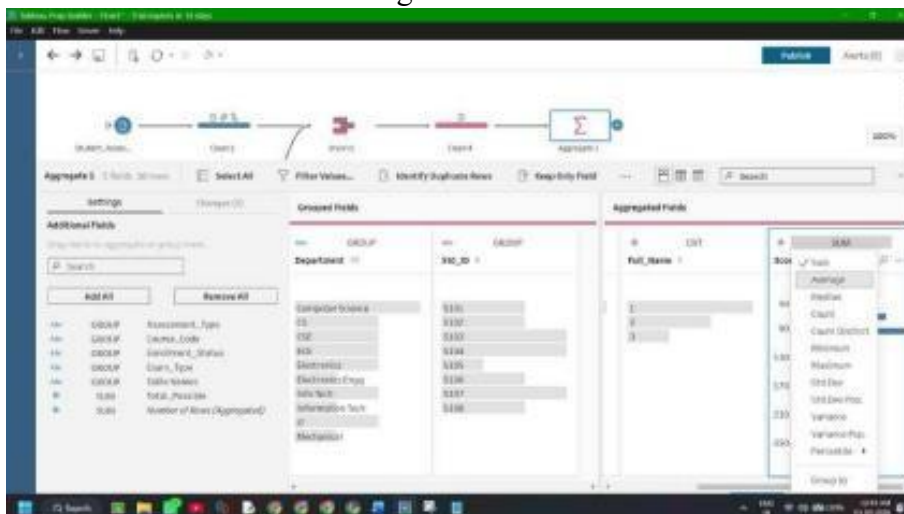


Step 4: Configure Aggregated Fields (Measures & Metrics)

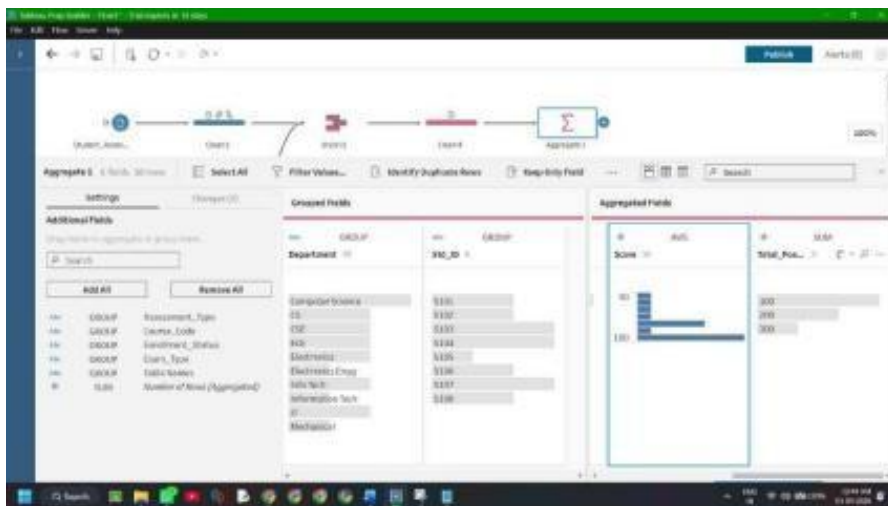
1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum).



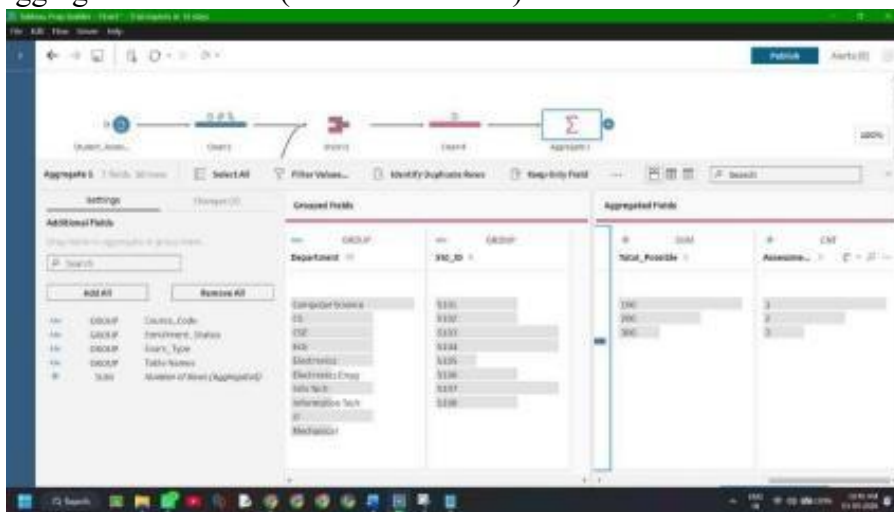
2. Average Score: Drag Marks_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.



2. Total Possible Marks: Drag Max_Marks into the Aggregated Fields box --> keep as Sum.



4. Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).



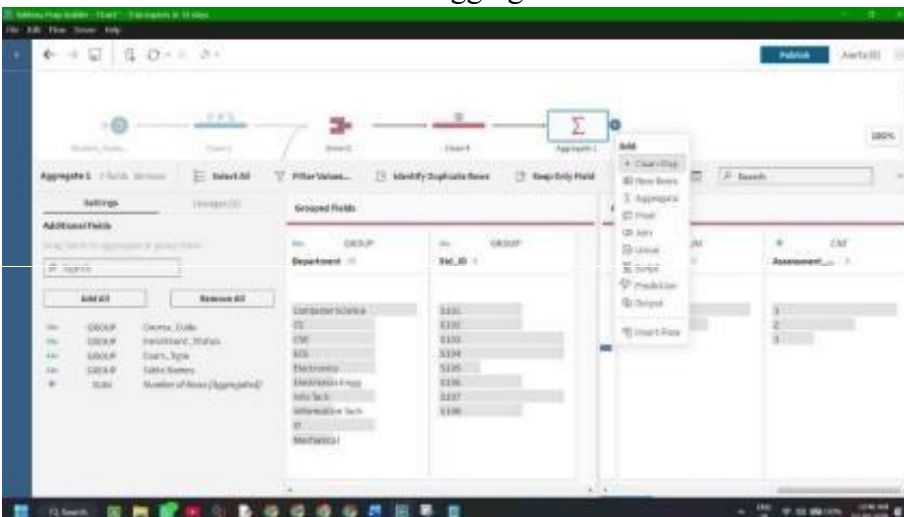
Step 5: Rename Aggregated Output Fields

In the Aggregated Fields panel, click on each field name to rename them cleanly:

- ☐ Marks_Obtained (Sum) --> Total_Marks_Obtained
- ☐ Marks_Obtained (Average) --> Average_Score
- ☐ Max_Marks (Sum) --> Total_Max_Marks
- ☐ Assessment_Type (Count) --> Assessments_Taken

Step 6: Add a Clean Step to Inspect Aggregated Output

1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean4).



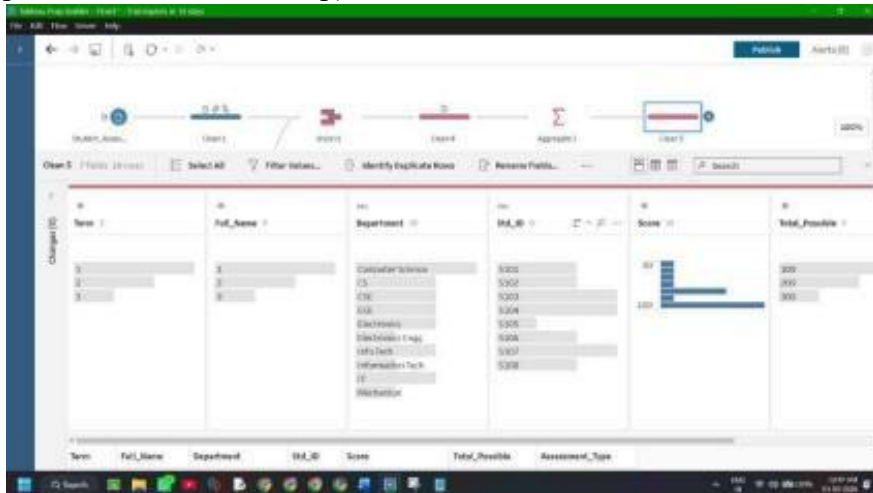
2. Look at the Profile Pane:

- o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

F |Apply Filters and Transformations Step

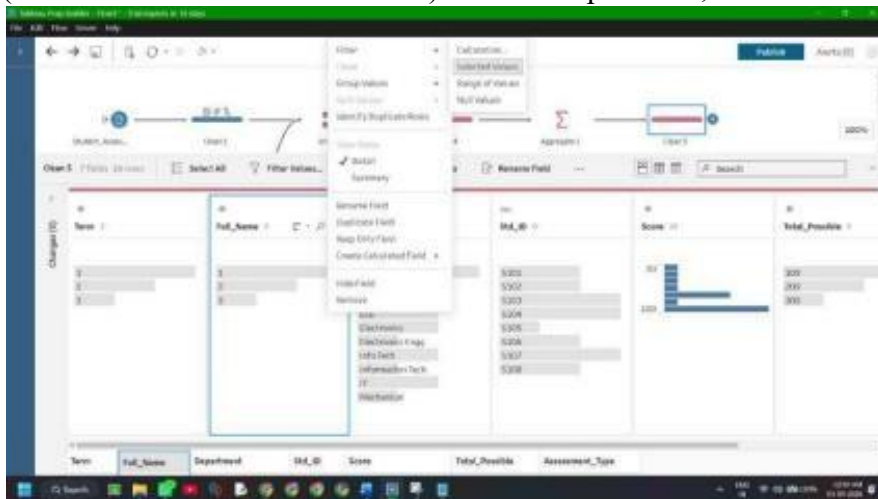
1: Open the Transformation Step (Clean 4)

1. Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas.
2. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

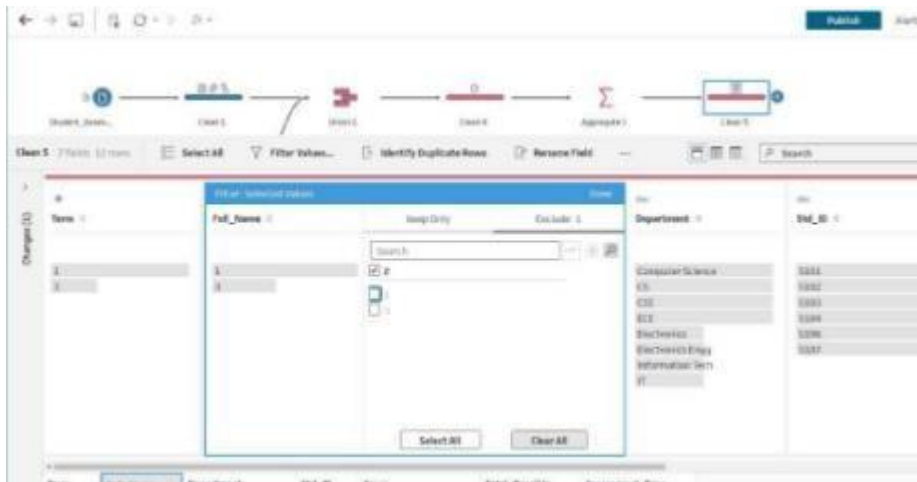


Step 2: Filter Out Inactive / Dropped Student Records

1. If Enrollment_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active).
2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



- o Exclude Dropped so only Active students remain in the active pipeline.



Step 3: Create the Normalized Term Percentage Metric

1. In the toolbar above the Profile cards, click Create Calculated Field. 2.

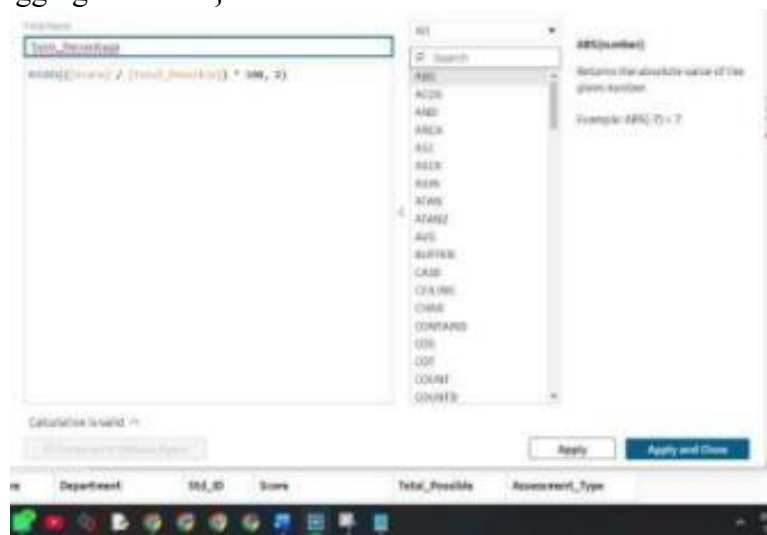
Field Name: Term_Percentage

3. Tableau Prep Formula:

Code snippet

$\text{ROUND}([\text{Total_Marks_Obtained}] / [\text{Total_Max_Marks}] * 100, 2)$

$\text{ROUND}([\text{Score}] / [\text{Total_Possible}] * 100, 2)$ {This formula can be used if you couldn't able to rename your aggregate fields.} 4. Click Save.



Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click Create Calculated Field. 2. Field Name:

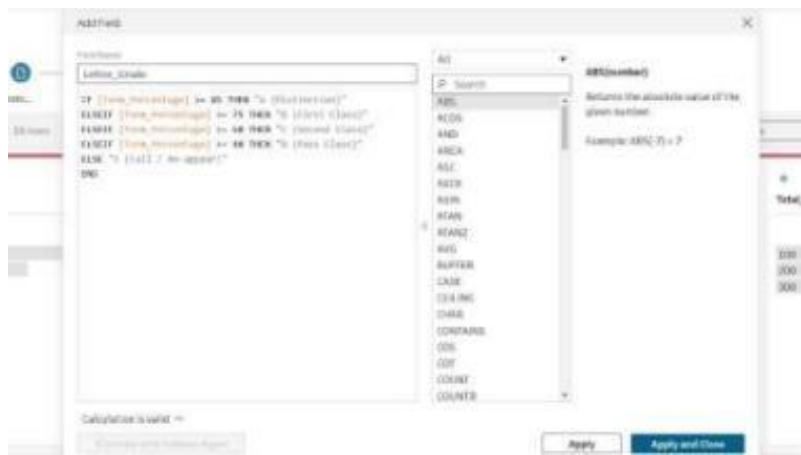
Letter_Grade

3. Tableau Prep Formula:

Code snippet

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END
```

3. Click Save.



Step 5: Create Academic Standing Compliance Flag 1.

Click Create Calculated Field.

2. Field Name: Academic_Standing

3. Tableau Prep Formula:

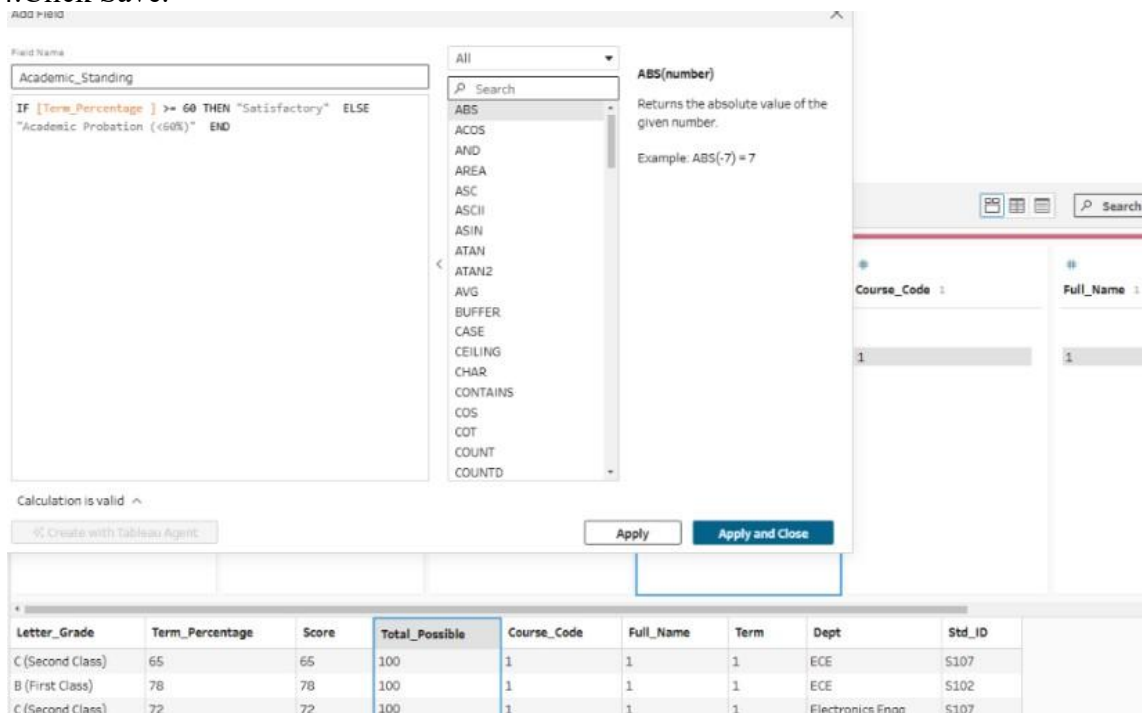
Code snippet

IF [Term_Percentage] >= 60 THEN "Satisfactory"

ELSE "Academic Probation (<60%)"

END

4. Click Save.



Step 6: Verify Transformed Profile Distributions

1. Look at the newly generated profile cards in Clean 4:

- o Term_Percentage: Displays a smooth numeric distribution between 0% and 100%.
- o Letter_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
- o Academic_Standing: Highlights students requiring academic counseling.



G| Create and automate data preparation flows

Step 1: Add an Output Step to the End of the Pipeline

1. On the Flow canvas, locate the final cleaning step (Clean 4).
2. Hover over the Clean 4 icon and click the + icon.
3. Select Output (represented by the download/save arrow icon).
4. An Output step node will appear connected directly to Clean 4.

Step 2: Configure Output Settings in the Bottom Pane

1. Click on the Output node to open its configuration panel below:
 - o Save output to: Select File (or Tableau Server / Tableau Cloud).
 - o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
 - o Location: Click Browse and choose the directory where you want the final dataset saved.
 - o File Name: Enter Student_Performance_Master_Cleaned.
 - o Write Option: Select Create table (or Append to table if updating existing records).

Step 3: Run the Data Prep Flow

1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon $\blacktriangleright$ directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating: "The flow ran successfully in X seconds."



Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As....

2. Name your workflow:

Plaintext

Student_Performance_Prep_Flow.tfl

3. (Automation Options in Production):

- o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep

Command Line Interface (CLI): Bash `tableau-prep-cli -t Student_Performance_Prep_Flow.tfl` o
 Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a
 scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are
 added.

The screenshot shows the Tableau Prep Desktop interface. On the left, a sidebar contains a 'Data' pane with a list of data sources, including 'Student_Performance_Master_1'. The main workspace displays a data flow diagram with steps like 'Extract', 'Join', 'Filter', and 'Aggregate'. Below the flow, a preview of the final dataset is shown, titled 'Save to Student_Performance_Master_Cleaned.csv'.

Letter_Grade	Term_Percentage	Academic_Standings	Term	Full_Name	Department	Stud_ID	Score
D (Poor Class)	86.5	Academic Probation (100%)	2	2	Info Tech	10000	91
A (Great! No repeat)	25.00	Academic Probation (100%)	1	3	Electronics	10001	90
A (Good Class)	90	Satisfactory	1	1	Information Tech	10002	90
A (Good Class)	89	Satisfactory	1	1	IT	10003	90
F (Fail! No repeat)	10.75	Academic Probation (100%)	2	2	Mechanical	10004	91
C (Good Class)	70	Satisfactory	1	1	IT	10005	90
C (Good Class)	68	Satisfactory	1	1	IT	10006	90
A (Good Class)	94	Satisfactory	1	1	IT	10007	90
A (Good! No repeat)	81.00	Academic Probation (100%)	2	2	Computer Science	10008	90
D (Poor Class)	47.5	Academic Probation (100%)	2	2	Info Tech	10009	90
A (Good Class)	90	Satisfactory	1	1	IT	10010	90
F (Fail! No repeat)	35	Academic Probation (100%)	2	2	Electronics	10011	90
A (Good Class)	75	Academic Probation	1	1	IT	10012	90

Step 5: Verify the Final Master Dataset in Tableau Desktop 1.

Open Tableau Desktop.

2. Connect to the exported Student_Performance_Master_Cleaned.csv (or .hyper).

3. Build the self-performance dashboard using the pre-calculated

